# Excel Function Cheat-Sheet

Quick reference: function name, syntax, and when to use it

| Function | Syntax | When to use |
|---|---|---|
| **Math & Statistics** | | |
| SUM | `=SUM(range)` | Adds numbers. Totals for a column of salaries or sales. |
| AVERAGE | `=AVERAGE(range)` | Mean of the numbers. Typical value of a data set. |
| MIN / MAX | `=MIN(range) =MAX(range)` | Smallest / largest value in a range. |
| COUNT / COUNTA | `=COUNT(range) =COUNTA(range)` | Counts numeric cells / all non-blank cells. |
| ROUND | `=ROUND(number, digits)` | Rounds to a set number of decimals (e.g. money to 2 places). |
| SUMIF / SUMIFS | `=SUMIF(range, criteria, sum_range)` | Adds only the rows that meet one (or many) conditions. |
| COUNTIF | `=COUNTIF(range, criteria)` | Counts cells that meet a condition. |
| AVERAGEIF | `=AVERAGEIF(range, criteria, avg_range)` | Averages only the rows that meet a condition. |
| **Logical** | | |
| IF | `=IF(test, if_true, if_false)` | Returns one result if a test is true, another if false. |
| AND / OR | `=AND(test1, test2) =OR(test1, test2)` | Combine several tests inside IF. |
| IFERROR | `=IFERROR(value, value_if_error)` | Replaces errors (#N/A, #DIV/0!) with a friendly result. |
| **Lookup & Reference** | | |
| VLOOKUP | `=VLOOKUP(lookup, table, col_num, FALSE)` | Finds a value in the first column and returns a value from a column to the right. Use FALSE for exact match. |
| XLOOKUP | `=XLOOKUP(lookup, lookup_range, return_range)` | Modern lookup; searches any direction, exact match by default. |
| INDEX / MATCH | `=INDEX(range, MATCH(lookup, lookup_range, 0))` | Flexible lookup that can return from any column. |
| **Text** | | |
| CONCAT / & | `=CONCAT(a, b) or =A2&" "&B2` | Joins text from several cells. |
| LEFT / RIGHT / MID | `=LEFT(text, n) =MID(text, start, n)` | Pulls part of a text string (IDs, codes). |
| LEN / TRIM | `=LEN(text) =TRIM(text)` | Counts characters / removes extra spaces. |
| UPPER / LOWER / PROPER | `=PROPER(text)` | Changes capitalization (e.g. clean up names). |
| **Date & Time** | | |
| TODAY / NOW | `=TODAY() =NOW()` | Current date / date and time (updates automatically). |
| DATEDIF | `=DATEDIF(start, end, "Y")` | Years ("Y"), months ("M") or days ("D") between dates. |
| YEAR / MONTH / DAY | `=YEAR(date)` | Extracts part of a date. |

**Cell references:** `C2` relative (changes when copied) | `$F$2` absolute (locked; shortcut F4) | `$F2` / `F$2` mixed.
**Example:** `=C2*$F$2` multiplies each salary by the one locked tax-rate cell. **Show formulas:** Ctrl + ` (grave accent).